

Abdul-Salem Byeibitkhan
Professor Sabina Adhikari
COMP 4353
05/01/2025

Dataset:

Source: [Fraud Detection Transactions Synthetic Dataset - Kaggle](#)

Rows: 50,000

Column: 21

Target Variable: Fraud_Label (0 = Not Fraud, 1 = Fraud)

GitHub Repository: https://github.com/abdulsal3m/COMP_4353_Final

Data Cleaning

The raw dataset was well-structured with no missing values. However, redundant columns like Transaction_ID and User_ID were dropped as they served no predictive purpose. The Timestamp column was converted to datetime64 format for potential temporal feature extraction. All categorical variables were encoded using One-Hot Encoding, and numeric fields were normalized using Min-Max Scaling.

Data Integration

The dataset was already quite rich in data given the more than 1,000,000 data points therefore, no additional data integration was necessary and the base indicators provided adequate context to yield reliable results.

Data Reduction

To simplify the model and reduce noise, the following data reduction methods were employed:

- Attribute Subset Selection: Dropped irrelevant columns like Transaction_ID, User_ID, and others based on domain knowledge.
- Histogram Analysis: Visualized feature distributions to spot outliers and skewed variables.
- Feature Importance Ranking using Random Forest: Helped confirm which attributes were critical for fraud prediction. Failed_Transaction_Count_7d and Risk_Score columns came out as the most notable and important columns.

Data Transformation

The raw dataset had numerous columns with high values, to prevent and counteract skewness, Min-Max scaling was implemented. This allowed it to bring all the features between 0 and 1, making it less prone to biased and inaccurate model outcomes and ensured that features on larger scales (e.g., Transaction_Amount vs. Risk_Score) didn't dominate the model during training.

Data Discretization

Binning was applied to high-range numeric columns to improve human interpretability and enable rule-based modeling in future applications, The following columns were binned into Low, Medium, High using pandas.cut():

- Transaction_Amount
- Account_Balance
- Risk_Score
- Transaction_Distance

This segmentation helped us explore trends across fraud categories and identify behavioral patterns.

Model Selection & Justification

Given the binary nature of the target variable (Fraud_Label), Classification models were chosen over Clustering.

Models Implemented:

- Random Forest: Chosen for its robustness, ensemble nature, and ability to handle both categorical and numerical variables well. Delivered excellent performance and clear feature importance rankings given its excellence in overfitting resistance.
- XGBoost: Implemented for benchmarking and focusing on misclassified cases, top choice for boosting accuracy through iterative training and ideal for tabular fraud datasets.
- Logistic Regression: Used as a baseline linear model. Helped highlight the strengths of ensemble methods by comparison.

Findings

Random Forest Results:

Accuracy: 100%

Top Features:

- Failed_Transaction_Count_7d: 53.9%
- Risk_Score: 40.9%
- Account_Balance, Transaction_Amount, and Card_Age: <1% each

Confusion Matrix:

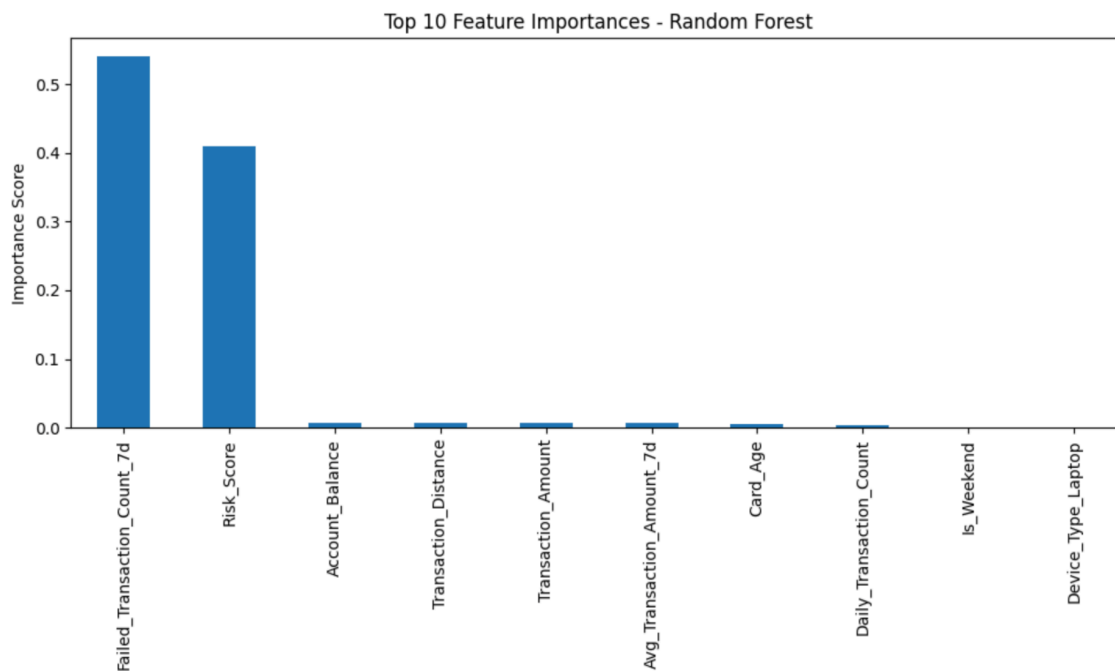
Confusion Matrix:

```
[[6787    0]
 [    0 3213]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	6787
1	1.00	1.00	1.00	3213
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000

Feature Importance:



XGBoost Results:

Accuracy: 100%

Top Features:

- Failed_Transaction_Count_7d
- Risk_Score
- Card_Type_Mastercard, Transaction_Type_Online, etc.

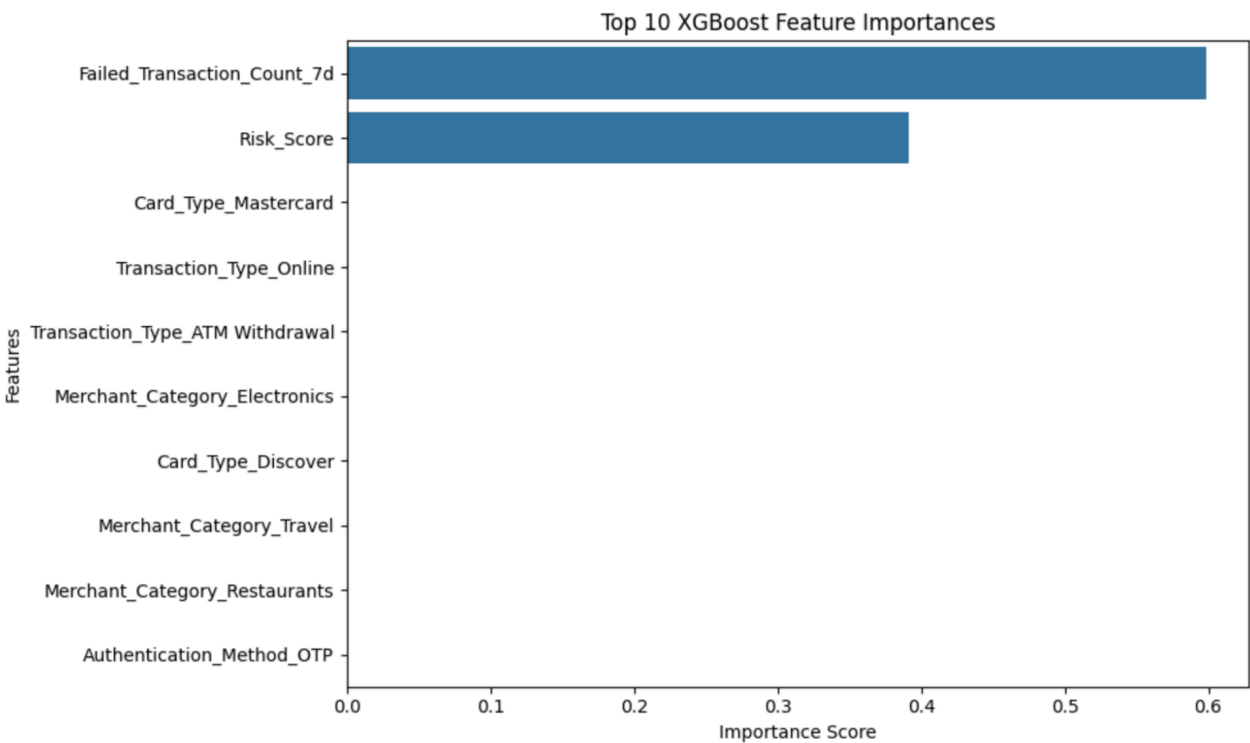
Confusion Matrix:

Confusion Matrix:
[[6783 4]
[4 3209]]

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	6787
1	1.00	1.00	1.00	3213
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000

Feature Importance:



Logistic Regression Results:

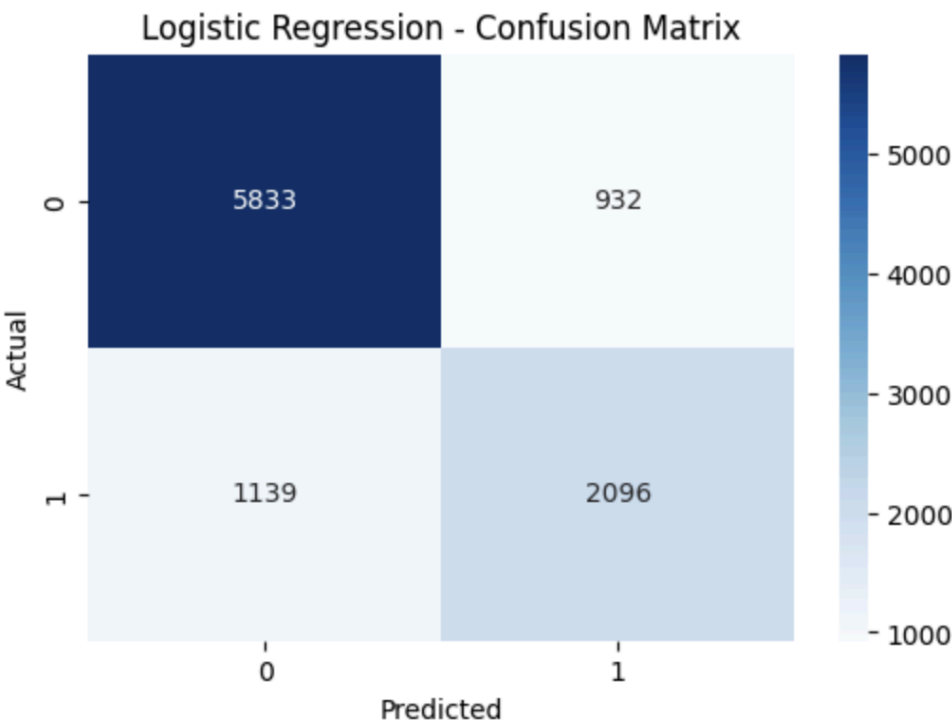
Accuracy: ~79%
Precision (Fraud Class): 0.69
Recall (Fraud Class): 0.65
F1 Score: 0.67

Confusion Matrix:

Confusion Matrix:
[[5833 932]
 [1139 2096]]

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.86	0.85	6765
1	0.69	0.65	0.67	3235
accuracy			0.79	10000
macro avg	0.76	0.76	0.76	10000
weighted avg	0.79	0.79	0.79	10000



Limitations

- Imbalanced Dataset: Although class distribution wasn't severely skewed, ensemble models likely benefited from the strong correlation between Risk_Score and fraud labels.
- Synthetic Nature: Since this is a simulated dataset, real-world fraud detection would require additional steps like temporal leakage checks and adversarial testing.